

SINAN, M

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Education

- **PhD in Biomedical Engineering** January, 2028. Warsaw, Poland
POLISH ACADEMY OF SCIENCES PAS,
Institute of Fundamental and Technological Research IPPT PAS,
Information and Biomedical Technologies TIB PAS
Expected: December 2028

Dissertation title: Computational analysis of multi-scale innate immune defences against bacterial pathogens
Research area: System biology, immunology, antibacterial innate immune responses
Supervisor: Prof. Dr. hab. Paweł Paszek
- **Master of Science in Mathematics**, July 2024. Chengdu, China
UNIVERSITY OF ELECTRONIC SCIENCE AND TECHNOLOGY OF CHINA, School of
Mathematical Sciences

Research area: Mathematical Biology
Supervisor: Prof. Dr. Leng Jinsong
- **BS in Mathematics**, November 2020. Swat, Pakistan
UNIVERSITY OF SWAT
Supervisor: Prof. Dr. Amir Khan

Awards and Achievements

- **Fully Funded University Scholarship**, 2021-2024
School of Mathematical Sciences, University of Electronic Science and Technology of China
- **Second rank in Outstanding Graduate Student Award**, 2022
International Student Office, University of Electronic Science and Technology of China
- **First prize in Academic Achievement Award**, 2022
School of Mathematical Sciences, University of Electronic Science and Technology of China

Research Experience

February 2025 – September 2027, Warsaw Poland

- **First Stage Researcher (R1)**,
POLISH ACADEMY OF SCIENCES PAS, Institute of Fundamental and Technological
Research IPPT PAS, Department of Biosystems and Soft Matter (ZBiMM), Division of Modelling
in Biology and Medicine (PMBM)

Selected Publications

Year – 2024

Naz, F., She, L., Sinan, M., & Shao, J. (2024). Enhancing Radar Echo Extrapolation by
ConvLSTM2D for Precipitation Nowcasting. *Sensors*, 24(2), 459.

Year – 2023

Sinan, M., & Alharthi, N. H. (2023). Mathematical analysis of fractal-fractional mathematical model
of COVID-19. *Fractal and Fractional*, 7(5), 358.

Sinan, M., Ansari, K. J., Kanwal, A., Shah, K., Abdeljawad, T., & Abdalla, B. (2023). Analysis of the
mathematical model of cutaneous leishmaniasis disease. *Alexandria Engineering Journal*, 72,
117-134.

Sinan, M., Leng, J., Shah, K., & Abdeljawad, T. (2023). Advances in numerical simulation with a
clustering method based on K-means algorithm and Adams Bashforth scheme for fractional
order laser chaotic system. *Alexandria Engineering Journal*, 75, 165-179.

Year – 2022

Rezapour, S., Etemad, S., Sinan, M., Alzabut, J., & Vinodkumar, A. (2022). A mathematical analysis
on the new fractal-fractional model of second-hand smokers via the power law type kernel:

numerical solutions, equilibrium points, and sensitivity analysis. *Journal of Function Spaces*, 2022

Shah, K., Sinan, M., Abdeljawad, T., El-Shorbagy, M. A., Abdalla, B., & Abualrub, M. S. (2022). A detailed study of a fractal-fractional transmission dynamical model of viral infectious disease with vaccination. *Complexity*, 2022.

Sinan, M., Ahmad, H., Ahmad, Z., Baili, J., Murtaza, S., Aiyashi, M. A., & Botmart, T. (2022). Fractional mathematical modeling of malaria disease with treatment & insecticides. *Results in Physics*, 34, 105220.

Sinan, M., Shah, K., Kumam, P., Mahariq, I., Ansari, K. J., Ahmad, Z., & Shah, Z. (2022). Fractional order mathematical modeling of typhoid fever disease. *Results in Physics*, 32, 105044.

Year – 2021

Ali, A., Khan, M. Y., Sinan, M., Allehiany, F. M., Mahmoud, E. E., Abdel-Aty, A. H., & Ali, G. (2021). Theoretical and numerical analysis of novel COVID-19 via fractional order mathematical model. *Results in physics*, 20, 103676.

Sinan, M., Ali, A., Shah, K., Assiri, T. A., & Nofal, T. A. (2021). Stability analysis and optimal control of COVID-19 pandemic SEIQR fractional mathematical model with harmonic mean type incidence rate and treatment. *Results in Physics*, 22, 103873.

Sinan, M., Shah, K., Khan, Z. A., Al-Mdallal, Q., & Rihan, F. (2021). On Semianalytical Study of Fractional-Order Kawahara Partial Differential Equation with the Homotopy Perturbation Method. *Journal of Mathematics*, 2021, 1-11

Talks

25-26 June 2026

“Fever modulates antibacterial innate immune response”, **19th Copernican Doctoral Seminar**, Nicolaus Copernicus University in Toruń, Poland

Research interest

Systems biology. Mathematical modelling. Signaling pathways. Data Science. Image segmentation, quantification and characterization. Machine Learning Algorithms

Skills

Language skill(s): Urdu. English

Software: MATLAB. Octave.

Programming Skills: Python (Pandas: multiIndexedDataFrames, Numpy, OOP). Image Processing. Data Processing. Deep Learning Algorithms.

Bioinformatics & Genomics: UCSC Genome Browser, In-Silico PCR, BLAT/BLAST sequence alignment. Correlating phenotypic data with genetic variants using browser annotation tracks.

Data Formats: FASTA, Pickle, Genetic Annotation Tracks (RefSeq, GENCODE).

Molecular Design: Primer design and specificity verification, identifying Promoter/TATA-box regions.

Analytical skill(s): Statistical Modeling, Predictive Analytics, Hypothesis Testing,

References

References are available on request